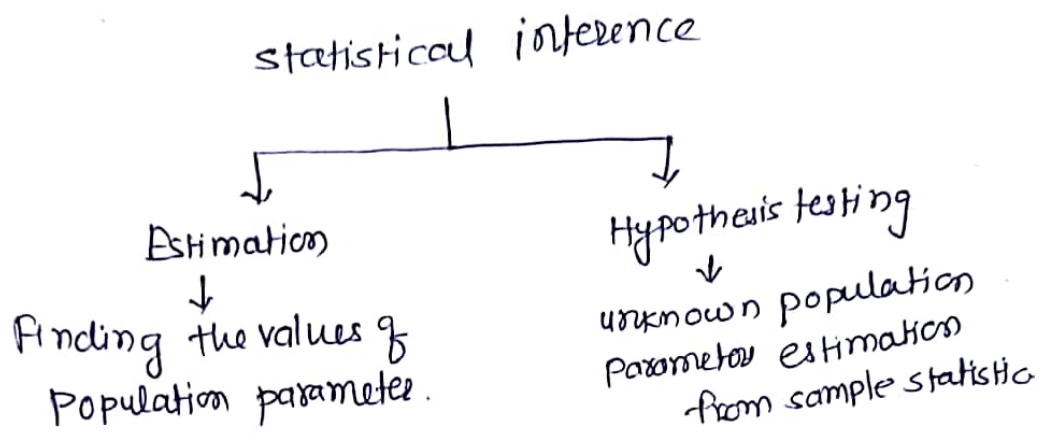


Unit 5: Inferential statistics: Hypothesis

①



Population: A collection of individuals/objects under study is called population. Two types

- 1) finite population :- contains finite no of individuals
- 2) infinite population - infinite number of population

Degrees of freedom :-

Degrees of freedom is number of independent values/variables that can vary in analysis without violating any constraints.

Test statistic

It is a number calculated from statistical test of hypothesis. This shows how well the observed data agree with expected distribution under the null hypothesis.

Statistical hypothesis

In many situations one may be interested in drawing certain conclusions about the population based on sample information. In taking such decisions certain assumptions are made. These assumptions are called 'Statistical hypothesis'.

Hypothesis is categorised into two types

1] simple hypothesis:

If the statistical hypothesis specifies the population completely, it is called as simple hypothesis.

2] composite hypothesis:

It is composed by many simple hypothesis and hence don't specify the population completely.

e.g. $H_0 = \mu = \mu_0$ is simple hypothesis

Where as

$$H_0: \mu < \mu_0$$

$$H_1: \mu > \mu_0$$

are composite hypothesis.

Test of statistical Hypothesis
(or Test of significance)

The method which one can decide whether to accept or reject a hypothesis is known as 'test of hypothesis' or hypothesis testing.

There are two types of hypothesis.

- 1) Null hypothesis
- 2) Alternate hypothesis

1. Null hypothesis:

1. It is hypothesis which is to be tested for possible rejection, under the assumption that is true.

2) Alternate hypothesis

the hypothesis that is different from the null hypothesis is called the alternate hypothesis.

e.g. to check

1. the average percentage of students in a college have mean 70 then.

$$H_0: \mu = 70 \text{ (Null hypothesis)}$$

vs

$$H_1: \mu \neq 70$$

or

$$H_1: \mu < 70$$

$$H_1: \mu > 70$$

} Alternate hypothesis.

Note:

1. Null hypothesis have an equal sign
2. the null and alternate hypothesis are always stated together.

Types of Error

Acceptance or rejection of null hypothesis depends on the sampling while making any conclusion about null hypothesis.

There are two types of errors.

Type I error:

Type I error happens when null hypothesis (H_0) is rejected even if it is true.

Type II error

Type II error happens when null hypothesis (H_0) is accepted when it is false.

	Reject H_0	Accept H_0
H_0 is true	Type I error	correct decision
H_0 is false	correct decision	Type II error

Level of significance

The probability of Type I error is known as level of significance (l.o.s.) It is denoted by α . It is known as size of the test.

The probability of Type I error α is fixed at low level generally at 5% or 1%, which means probability of accepting correct hypothesis is 95%.

$$\begin{aligned}\alpha &= P[\text{Reject } H_0 \mid H_0 \text{ is true}] \\ &= P[\text{Reject } H_0 \mid H_0] \\ &= P(\text{Type I error})\end{aligned}$$

Power of Test

Probability of Type II error is denoted by β
i.e. β is probability of accepting wrong null hypothesis. Therefore $1-\beta$ gives the probability of rejecting null hypothesis when it is false.
The probability $(1-\beta)$ is called Power of the test.

$$\begin{aligned}\beta &= [\text{Accept } H_0 \mid H_0 \text{ is false}] \\ &= [\text{Accept } H_0 \mid H_1 \text{ is true}] \\ &= [\text{Accept } H_0 \mid H_1] \\ &= P [\text{Type II error}]\end{aligned}$$

Remarks:

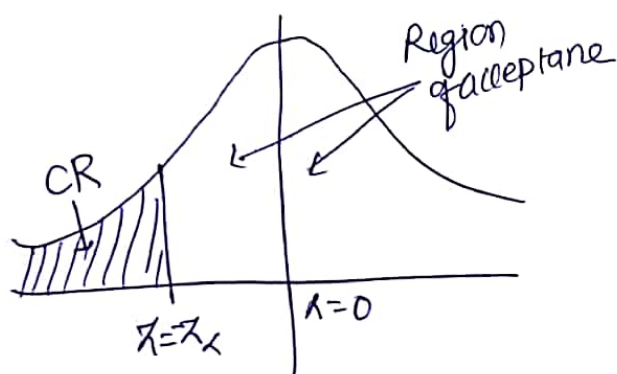
- 1) Smaller the value of β , larger the value of $1-\beta$ i.e. chances of rejecting false hypothesis are more. Thus test will work quite well.
- 2) Low value of $1-\beta$ (or close to zero) means that test is working very poorly.

critical Region

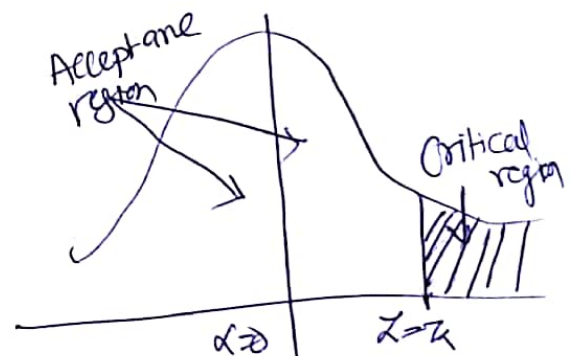
Hypothesis testing involves determining the region such that hypothesis is rejected if the sample points fall in the region and it will be accepted if the sample points are outside the region. The region of rejection is called 'critical region'. It is denoted by W , acceptance region is denoted by $\bar{W}$.

Remark

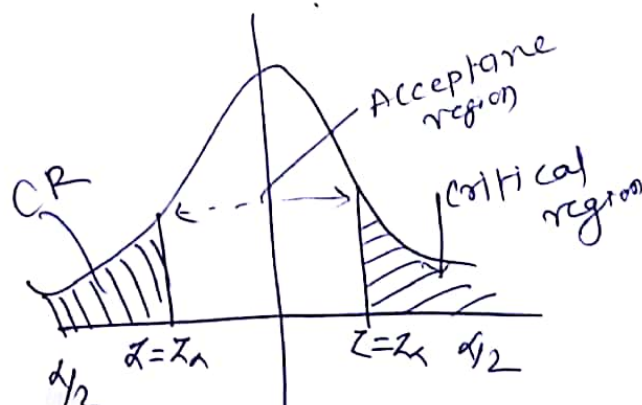
1. If the critical region lies on the one side/ tail of distribution of test value then such tests are called one tailed or one sided test.
2. If the critical region lies on the left side of the distribution then it is known as left tailed test.
3. If the CR lies on the right side of the distribution then it is known as right tailed test.
4. If CR lies on both tails of distribution then such tests are called two tailed tests.



Left tailed test



Right tailed test



Two tailed test

$$\alpha = \frac{3}{16}$$

β = probability of type II error

$$= P[\text{Accept } H_0 | H_1]$$

$$= P[X \leq 3 | P = \frac{3}{4}]$$

$$= 1 - P[X \geq 4 | P = \frac{3}{4}]$$

$$= 1 - \left\{ P[X=4 | P = \frac{3}{4}] + P[X=5 | P = \frac{3}{4}] \right\}$$

$$= 1 - \left\{ {}^5C_4 \left(\frac{3}{4}\right)^4 \left(1 - \frac{3}{4}\right)^{5-4} + {}^5C_5 \left(\frac{3}{4}\right)^5 \left(1 - \frac{3}{4}\right)^{5-5} \right\}$$

$$= 1 - \frac{47}{128}$$

$$\text{Then power of test} = 1 - \beta = 1 - \frac{47}{128} = \frac{81}{128}$$

Q. 2

Let P is the probability that a given die shows even number. to test $H_0: P = \frac{1}{2}$

vs $H_1: P = \frac{1}{3}$. following procedure is adopted. toss the die twice and accept H_0 if both times it shows even number. Find the probabilities of Type I and Type II error.

Soln

$$H_0: P = \frac{1}{2} \quad \text{vs} \quad H_1: P = \frac{1}{3}$$

$$n = 2$$

Let x denote the even no.

by Binomial distribution

$$P(X = x) = {}^nC_x P^x (1-P)^{n-x}$$

Q. Let P be the probability that a coin will fall head in a single toss in order to test $H_0: P = \frac{1}{2}$ vs $P = \frac{3}{4}$. the coin is tossed 5 times and H_0 is rejected if more than 3 heads are obtained. Find the probability of Type I error and power of test

Soln: Given $H_0: P = \frac{1}{2}$ vs $H_1: P = \frac{3}{4}$

$$n = 5$$

Let x denote the no of heads in n tosses then by binomial distribution

$$P(X=x) = {}^nC_x P^x q^{n-x}$$

$$\text{Q } n=5 \quad q=1-P$$

$$P(X=x) = {}^5C_x P^x (1-P)^{5-x} \quad \text{--- (1)}$$

Given that H_0 is rejected if more than 3 heads are obtained.

$\therefore$ critical region is given by

$$W = \{x / x \geq 4\}$$

$$\bar{W} = \{x / x \leq 3\}$$

α = probability of Type I error

$$= P[\text{Reject } H_0 / H_0]$$

$$= P[x \geq 4 / P = \frac{1}{2}]$$

$$= P[x=4 / P = \frac{1}{2}] + P[x=5 / P = \frac{1}{2}]$$

$$= {}^5C_4 \left(\frac{1}{2}\right)^4 \left(\frac{1}{2}\right)^{5-4} + {}^5C_5 \left(\frac{1}{2}\right)^5 \left(\frac{1}{2}\right)^{5-5}$$

we reject H_0 if no of times, that even number is shown is less than 2.

C.R is

$$W = \{x \mid x < 2\}$$

$$\bar{W} = \{x \mid x = 2\}$$

$$\alpha = P[\text{Reject } H_0 \mid H_0 \text{ is true}]$$

$$= P[X < 2 \mid P = \frac{1}{2}]$$

$$= 1 - P[X = 2 \mid P = \frac{1}{2}]$$

$$= 1 - {}^2C_2 \left(\frac{1}{2}\right)^2 \left(\frac{1}{2}\right)^{2-2}$$

$$= 1 - \frac{1}{4} = \frac{3}{4}$$

$$\boxed{\alpha = \frac{3}{4}}$$

$$\beta = P[\text{Accept } H_0 \mid H_1]$$

$$= P[X = 2 \mid P = \frac{1}{3}]$$

$$= {}^2C_2 \left(\frac{1}{3}\right)^2 \left(1 - \frac{1}{3}\right)^{2-2}$$

$$\beta = \frac{1}{9}$$

$$\begin{aligned} \text{power of test} &= 1 - \beta \\ &= 1 - \frac{1}{9} \\ &= \frac{8}{9} \end{aligned}$$

Test of significance

depending on the sample size. we categorise testing of significance into two types.

1. Test of significance for large samples
(sample size $n \geq 30$)
2. Test of significance for small samples
(sample size $n < 30$)

Test of significance of small samples.

When the sample size is less than 30 then the sample is called small sample.

- 1) Student's t-test (t-test)
- 2) F-test.

1] Student's t-test — t-test assumes

- 1) Normal distribution
- 2) population s.d. unknown
- 3) small sample ($n < 30$)

for mean of one samples

for mean of two samples

If $X_1, X_2, \dots, X_n$ is the random sample from the normal population $N(\mu, \sigma^2)$ then t-statistic is defined as

$$t = \frac{\bar{X} - \mu}{s/\sqrt{n}}$$

where, $\bar{X}$ - sample mean
 μ - population mean
 s - s.d. of the sample, n - sample size

$$\bar{X} = \frac{\sum X_i}{n}, \quad s = \sqrt{\frac{\sum (X_i - \bar{X})^2}{n-1}}, \text{ with degrees of freedom } n-1$$

Q.1 A random sample of 16 newcomers gave mean of 1.67 m and s.d of 0.16 m. The mean height of the students of the previous year known to be 1.600 m. Is the mean height of the newcomers significantly different from the main height of the students population of the previous year? Given $t_{0.05, 15} = 2.13$

Solⁿ: H_0 : there is no significant difference between mean height of the newcomers and the mean height of the student population of previous year.

ie. $H_0: \mu = 1.60$

vs
 $H_1: \mu \neq 1.60$

here, $n = 16$, $\bar{X} = 1.67$, $\mu = 1.60$
 $s = 0.16$

$$t = \frac{\bar{X} - \mu}{\frac{s}{\sqrt{n}}} = \frac{1.67 - 1.60}{0.16 / \sqrt{16}}$$

$$= 1.75$$

degrees of freedom $= n - 1 = 16 - 1 = 15$

$$|t| = 1.75 < t_{0.05, 15} = 2.13$$

$\therefore H_0$ accepted at 5% l.o.s.

Q.2

Suppose that sweets are sold in packages of fixed weight of the content. The procedure of packages is interested in testing the average weight of content in packages in 1 kg. hence a random sample of 12

Packages is drawn and their contents found (in kg) are as follows: 1.05, 1.01, 1.04, 0.98, 0.96, 1.01, 0.97, 0.99, 0.98, 0.95, 0.97, 0.95. Using the above data what should he conclude about the average. (Given $t_{0.05, 11} = 2.201$)

Soln: $H_0: \mu = 1$
vs $H_1: \mu \neq 1$.

$$n = 12$$

$$\bar{x} = \frac{\sum x_i}{n} = \frac{11.86}{12} = 0.9883$$

$$\sum (x_i - \bar{x})^2 = 0.011967$$

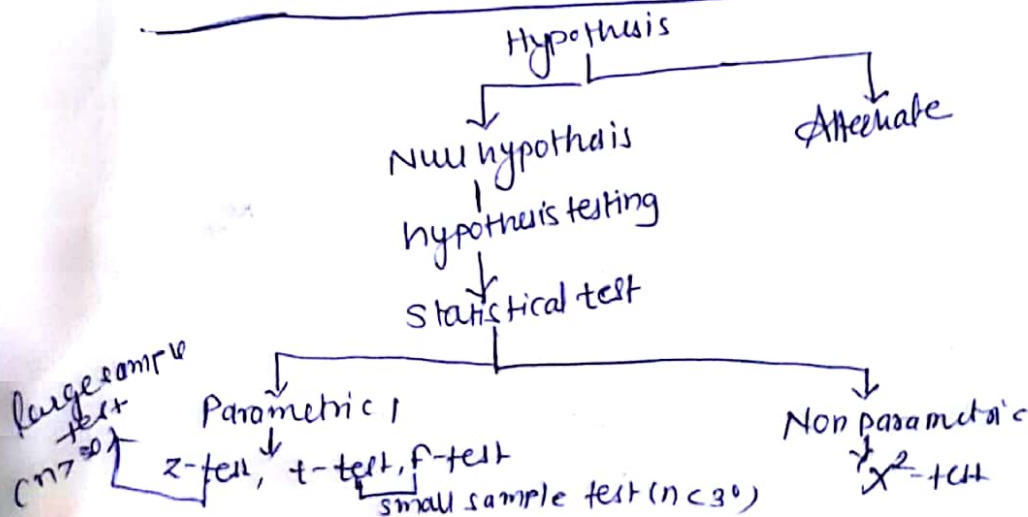
$$s = \sqrt{\frac{\sum (x_i - \bar{x})^2}{n-1}} = \sqrt{\frac{0.011967}{11}} = 0.032983$$

$$t = \frac{\bar{x} - \mu}{s/\sqrt{n}} = \frac{0.9883 - 1}{0.032983/\sqrt{12}} = -1.228815$$

$$|t| = 1.228815 < t_{0.05, 11} = 2.201$$

H_0 accepted at 5% level of S.

$\therefore$ The average weight of contents of Packages taken is 1 kg.



two samples

Q.3 A random sample of size 16 has 53 as mean. the sum of squares of the deviation from mean is 135. can this sample be regarded as taken from the population having 56 as mean?
(Given $t_{0.05} = 1.753$)

soln: $H_0: \mu = 56, H_1: \mu \neq 56$

Given $\bar{x} = 53, \mu = 56, n = 16,$
 $\sum (x_i - \bar{x})^2 = 135$

$$S = \sqrt{\frac{\sum (x_i - \bar{x})^2}{n-1}} = \sqrt{\frac{135}{15}} = 3$$

$$t = \frac{53-56}{3/\sqrt{16}} = -4$$

$|t| = 4$ degrees of freedom $= 16-1 = 15$

Since $|t| = 4 > t_{0.05} = 1.753$

i.e. calculated value of t is more than the table value.

$\therefore$ the hypothesis is rejected.

t-test for difference of means of two

Sample Q

two samples $x_1, x_2, \dots, x_{n_1}$, $y_1, y_2, \dots, y_{n_2}$ of sizes n_1, n_2 have been drawn from two normal populations with mean μ_1 and μ_2 respectively

t test is given by

$$t = \frac{(\bar{x} - \bar{y})}{S \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}}$$

NOTE: $\bar{x}_1 = \frac{\sum x_i}{n_1}$ $\bar{x}_2 = \frac{\sum y_i}{n_2}$

$$S^2 = \frac{\sum_{i=1}^{n_1} (x_i - \bar{x})^2 + \sum_{i=1}^{n_2} (y_i - \bar{y})^2}{n_1 + n_2 - 2}$$

with degrees of freedom $n_1 + n_2 - 2$

Note 1 If two sample S.D s_1, s_2 are given

$$S^2 = \frac{n_1 s_1^2 + n_2 s_2^2}{n_1 + n_2 - 2}$$

2. If $n_1 = n_2 = n$, $t = \frac{\bar{x} - \bar{y}}{\sqrt{\frac{s_1^2 + s_2^2}{n-1}}}$

3. Let $x_1, x_2, \dots, x_n$ and $x'_1, x'_2, \dots, x'_n$ be the values of two random sample from two different populations $N(\mu_1, \sigma^2)$, $N(\mu_2, \sigma^2)$ having μ_1 and μ_2 but the same S.D. then t-test statistic is

$$t = \frac{\bar{x} - \bar{x}' - (\mu_1 - \mu_2)}{S \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}} \text{ with } n_1 + n_2 - 2 \text{ d.o.f.}$$

Q.1

Fertilizers A and B are tried respectively on 10 and 8 randomly chosen experimental plots. The yields in the plots were as given below. Test whether the effects of the fertilizers are reflected in the mean yields.

Fertilizer	Yields									
A	8.0	7.6	8.2	7.8	8.3	8.4	8.2	7.8	7.7	8.0
B	7.4	8.1	7.6	8.1	7.5	7.6	7.3	7.2		

Soln.

$$H_0: \mu_1 = \mu_2$$

$$H_1: \mu_1 \neq \mu_2$$

$$\text{here } n_1 = 10, \bar{X}_1 = 8.0, \sum (x_i - \bar{X}_1)^2 = 0.66$$

$$n_2 = 8, \bar{X}_2 = 7.6, \sum (x_i - \bar{X}_2)^2 = 0.80$$

$$S^2 = \frac{0.66 + 0.80}{16} = 0.09125$$

$$t = \frac{8.0 - 7.6}{\sqrt{0.09125} \sqrt{\frac{1}{10} + \frac{1}{8}}} = \frac{\bar{X}_1 - \bar{X}_2}{S \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}}$$

$$= 2.79.$$

$$\text{Degrees of freedom} = n_1 + n_2 - 2 = 16$$

Q.2. ^{of} Sample size 10 and 14 were taken from two normal population with s.d. 3.5 and 5.2. The sample means were found to be 20.3 and 18.8. Test whether the means of the two populations are at the same level.

Soln.

$$H_0: \mu_1 = \mu_2$$

$$H_1: \mu_1 \neq \mu_2$$

$$\bar{X}_1 = 20, n_1 = 10, s_1 = 3.5$$

$$\bar{X}_2 = 18.6, n_2 = 14, s_2 = 5.2$$

$$s^2 = \frac{n_1 s_1^2 + n_2 s_2^2}{n_1 + n_2 - 2} = \frac{10(3.5)^2 + 14(5.2)^2}{10 + 14 - 2} = 22.775$$

$$\therefore s = 4.772$$

$$\therefore t = \frac{\bar{X}_1 - \bar{X}_2}{s \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}} = \frac{20.3 - 18.6}{4.772 \left(\sqrt{\frac{1}{10} + \frac{1}{14}} \right)} = 0.8604$$

$$|t| = 0.8604 < t_{0.05, 22} = 2.07$$

$\therefore H_0$ is accepted.

✓

F test

Let $x_1, x_2, \dots, x_n$ and $x'_1, x'_2, \dots, x'_n$ are two independent random samples from two normal population $N(\mu_1, \sigma_1^2)$ and (μ_2, σ_2^2) respectively then the statistics of F-distribution is

$$F = \frac{s_1^2}{s_2^2} \quad \text{where } s_1^2 > s_2^2$$

$$\text{Where, } s_1^2 = \frac{\sum (x_i - \bar{x}_1)^2}{n_1 - 1} \quad \text{and} \quad s_2^2 = \frac{\sum (x'_i - \bar{x}'_1)^2}{n_2 - 1}$$

n_1 - ~~size~~ & n_2 : sizes of two samples with variances s_1^2 and s_2^2

Remark

1. In F Test, population variances are equal
i.e. $\sigma_1^2 = \sigma_2^2$
2. degrees of freedom $(n_1 - 1, n_2 - 1)$
3. Test result: If calculated value of F exceeds $F_{0.05}$ for $(n_1 - 1, n_2 - 1)$ d.o.f from the given F-table. we conclude that there is significant difference between the variances of two population
4. F test is applied when the sample drawn from the same population or the population with same variances.

Q. In two independent samples of size 8 and 10. the sum of squares deviations of the sample values from the respective sample means were 84.4 and 102.6. Test whether the difference of variances of the population is significant or not. (Given $F_{0.05} = 3.29$ at d.o.f (7,9),

Solⁿ: H_0 : there is no significant difference between Population variances $\sigma_1^2 = \sigma_2^2$

vs
 H_1 : $\sigma_1^2 \neq \sigma_2^2$

for sample 1, $n_1 = 8$, $\sum (x_i - \bar{x}_1)^2 = 84.4$

for sample 2 $n_2 = 10$ $\sum (x_i' - \bar{x}_2')^2 = 102.6$

$$S_1^2 = \frac{\sum (x_i - \bar{x}_1)^2}{n_1 - 1} = \frac{84.4}{8 - 1} = \frac{84.4}{7} = 12.057$$

$$S_2^2 = \frac{\sum (x_i' - \bar{x}_2')^2}{n_2 - 1} = \frac{102.6}{9} = 11.4$$

$$F = \frac{S_1^2}{S_2^2} \quad (\text{with } \sigma_1^2 = \sigma_2^2 \text{ crossed out})$$

$$= \frac{12.057}{11.4}$$

$$= 1.0576$$

$$\therefore |F| = 1.057 < F_{0.05} = 3.29$$

$\Rightarrow H_0$ is accepted.

$\therefore$ there is no significant difference between the variance of two population.

Q.2

the following table gives the number of units produced per day by 2 workers A and B for some days. can we say that worker A is more stable than worker B. Use $\alpha = 0.05$. (Given $F_{0.05, 7, 5} = 4.88$)

Worker A	40	30	38	41	38	35		
Worker B	39	38	41	33	32	49	49	34

Solⁿ:

H_0 : the stability of both workers is same

$$\sigma_1^2 = \sigma_2^2$$

H_1 : worker A is more stable than worker B

$$n_1 = 6, n_2 = 8$$

$$\bar{x}_1 = \frac{\sum x_i}{n_1} = \frac{222}{6} = 37$$

$$\bar{x}_2 = \frac{\sum x_i}{n_2} = \frac{315}{8} = 39.38$$

$$s_1^2 = \frac{\sum (x_i - \bar{x}_1)^2}{n_1 - 1} = \frac{80}{6-1} = 16$$

(Note for worker A $\sum (x_i - \bar{x}_1)^2 = 80$)

for worker B $\sum (x_i - \bar{x}_2)^2 = 313.99$

$$s_2^2 = \frac{\sum (x_i - \bar{x}_2)^2}{n_2 - 1} = \frac{313.99}{8-1} = 44.86$$

F test statistic

$$F = \frac{s_2^2}{s_1^2} = \frac{44.86}{16} = 2.80 \quad (\because s_2^2 > s_1^2)$$

Degrees of freedom, $n_2 - 1, n_1 - 1 = 8 - 1, 6 - 1, (7.5)$

$$|F| = 2.80 < F_{0.05, (7.5)} = 4.88$$

$\therefore H_0$ is accepted

i.e. the stability of both the workers is same.

Q3 from the following information. calculate F statistics and also state degrees of freedom.

	sample 1	sample 2
size (n)	10	13
Mean ($\bar{x}$)	50	52
sum of square of deviations $\sum (x_i - \bar{x})^2$ from mean	144	264

Soln:

$$S_1^2 = \frac{\sum (x_i - \bar{x}_1)^2}{n_1 - 1} = \frac{144}{10 - 1} = 16$$

$$S_2^2 = \frac{\sum (x_i - \bar{x}_2)^2}{n_2 - 1} = \frac{264}{13 - 1} = 22$$

$$\therefore F = \frac{S_2^2}{S_1^2} \quad (\because S_2^2 > S_1^2)$$

$$= \frac{22}{16} = 1.38$$

$$\text{degrees of freedom} = (n_2 - 1, n_1 - 1) = (13 - 1, 10 - 1) = (12, 9)$$

Remark: when readymade data is given instead of tabular long data. we use

$$S_1^2 = \frac{n_1 s_1^2}{n_1 - 1}$$

$$S_1^2 = \frac{\sum x_i^2 - n_1 (\bar{x}_1)^2}{n_1 - 1}$$

$$S_2^2 = \frac{n_2 s_2^2}{n_2 - 1}$$

$$S_2^2 = \frac{\sum x_i^2 - n_2 (\bar{x}_2)^2}{n_2 - 1}$$

Q. Find the F-statistics from the following data

Sample	size	Total observation Σx	Sum of squares of observation Σx^2
1	8	9.6	61.52
2	11	16.5	73.26

Test

Soln

$$\bar{x}_1 = \frac{\Sigma x_i}{n_1} = \frac{9.6}{8} = 1.2$$

$$\bar{x}_2 = \frac{\Sigma x_i}{n_2} = \frac{16.5}{11} = 1.5$$

$$S_1^2 = \frac{\Sigma x_i^2 - n_1(\bar{x}_1)^2}{n_1 - 1} = \frac{61.52 - 8(1.2)^2}{8 - 1} = 7.14$$

$$S_2^2 = \frac{\Sigma x_i^2 - n_2(\bar{x}_2)^2}{n_2 - 1} = \frac{73.26 - 11(1.5)^2}{11 - 1} = 4.85$$

$\therefore$ F test statistics

$$F = \frac{S_2^2}{S_1^2} = \frac{4.85}{7.14} = 0.68$$

Q. From the following information, calculate F-statistic and also its degrees of freedom.

	sample 1	sample 2
size (n)	22	16
S.D (s)	2.9	3.8

Soln

$$S_1^2 = \frac{n_1 s_1^2}{n_1 - 1} = \frac{22(2.9)^2}{22 - 1} = 8.81$$

$$S_2^2 = \frac{n_2 s_2^2}{n_2 - 1} = \frac{16(3.8)^2}{16 - 1} = 15.4$$

$$F = \frac{S_1^2}{S_2^2} = \frac{8.81}{15.4} = 0.57 \text{ dof } (15, 21)$$

Test of significance for large samples

If the sample size $n > 30$, the sample is taken as large sample.

Following are the important tests to test the significance of large samples.

1. Testing of significance for single ~~pop~~ proportion.
2. Testing of significance for difference of proportion.
3. Testing of significance for single mean.
4. Testing of significance of difference of means

Test of significance for single mean.

Let $x_1, x_2, \dots, x_n$ be a random sample of a size n from a large population of size N and mean μ and variance σ^2 .

The test statistic is

$$Z = \frac{\bar{X} - \mu}{\sigma / \sqrt{n}}$$

where $\bar{X}$ - mean of a sample

μ - mean of population

σ - S.D of the population

n - sample size

Q. A normal population has mean 6.8 and s.d. of 1.5. A sample of 400 members gave a mean of 6.75. Is the difference significant? (Given, $Z_{0.05} = 1.96$)

Soln: Given $\mu = 6.8$, $\sigma = 1.5$, $\bar{X} = 6.75$,
 $n = 400$

H_0 : there is a significant difference between sample mean & population mean

$$\text{i.e. } \mu = \bar{X} \quad \mu = 6.8$$

vs
 H_1 : $\mu \neq \bar{X}$ i.e. $\mu \neq 6.8$

$$Z = \frac{\bar{X} - \mu}{\sigma / \sqrt{n}} = \frac{6.75 - 6.8}{1.5 / \sqrt{400}} = -0.67$$

$$|Z| = 0.67 < Z_{0.05} = 1.96 \text{ at } 5\% \text{ l.o.s.}$$

$\therefore H_0$ is accepted.

Q.2 The height of college students in the city are normally distributed with s.d. 6 cm. A sample of 1000 students has mean height 158 cm. Test the hypothesis mean height of college students in the city is 160 cm.

Soln Given (Z_{α} at 5% l.o.s = 1.96)

$$n = 1000, \sigma = 6, \mu = 160, \bar{X} = 158$$

$$Z = 10.54 \Rightarrow H_0 \text{ is rejected}$$

Note: For large samples, if σ is unknown

$$Z = \frac{\bar{X} - \mu}{s / \sqrt{n}} \quad s = \text{s.d. of the sample}$$

#2) Test of significance for two means (for large samples)

Let $\bar{X}_1$ be the mean of sample size n_1 from the population with μ_1 and variance σ_1^2 . Let $\bar{X}_2$ be the mean of an independent sample of size n_2 from another population with mean μ_2 and variance σ_2^2 .

test statistic is

$$Z = \frac{\bar{X}_1 - \bar{X}_2}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}}$$

Note:

if σ_1 and σ_2 are unknown

$$Z = \frac{\bar{X}_1 - \bar{X}_2}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}$$

s_1, s_2 - S.D of sample I & II

Case I: If $\sigma_1 = \sigma_2 = \sigma$

$$Z = \frac{\bar{X}_1 - \bar{X}_2}{\sigma \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}}$$

Case II: If $\sigma_1 = \sigma_2$ & u not known.

$$\sigma^2 = \frac{n_1 s_1^2 + n_2 s_2^2}{n_1 + n_2}$$

Q.1 for sample I, $n_1 = 1000$, $\sum x = 49000$,

$$\sum (x - \bar{x})^2 = 7,84,000$$

for sample II: $n_2 = 1500$, $\sum x = 70500$, $\sum (x - \bar{x})^2 = 2400000$

Discuss the significant difference between the mean score.

Soln

H_0 : there is no significant difference between the sample means

$$\bar{X}_1 = \bar{X}_2$$

$$H_1: \bar{X}_1 \neq \bar{X}_2$$

$$S_1^2 = \frac{\sum (x_1 - \bar{x}_1)^2}{n_1} = \frac{784000}{1000} = 784$$

$$S_2^2 = \frac{\sum (x_2 - \bar{x}_2)^2}{n_2} = \frac{2400000}{1500} = 1600$$

$$\bar{X}_1 = \frac{\sum x_i}{n_1} = \frac{49000}{1000} = 49$$

$$\bar{X}_2 = \frac{\sum x_{i2}}{n_2} = \frac{70500}{1500} = 47$$

Test statistic

$$Z = \frac{\bar{X}_1 - \bar{X}_2}{\sqrt{\frac{S_1^2}{n_1} + \frac{S_2^2}{n_2}}} = \frac{49 - 47}{\sqrt{\frac{784}{1000} + \frac{1600}{1500}}} = 1.470$$

$$|Z| = 1.47 < Z_{0.05} = 1.96$$

$\therefore H_0$ is accepted

there is no significant difference between the sample means.

Q.2 From the given table below. Intelligence test of two groups of boys and girls gave the following results. Examine the difference is significant.

	Mean	S.D.	Size
Girls	70	10	70
boys	75	11	110

Soln:

$$H_0: \bar{X}_1 = \bar{X}_2$$

$$H_1: \bar{X}_1 \neq \bar{X}_2$$

$$Z = \frac{\bar{X}_1 - \bar{X}_2}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}} = \frac{70 - 75}{\sqrt{\frac{(10)^2}{70} + \frac{(11)^2}{110}}} = -3.144$$

$$|Z| = 3.144 > Z_{\alpha} = 1.96 \text{ at } 5\% \text{ L.O.S.}$$

H_0 is rejected.

there is a significant difference of the intelligence of boys and girls.

Test of significance of single proportion

(large samples)

We use this test to check the significant difference between proportion of sample and the population.

test statistic is

$$Z = \frac{p - P}{\sqrt{PQ/n}}$$

where, p - proportion of success in sample.

$$p = \frac{X}{n}$$

X : no of success in n independent trials

P = population proportion

$$Q = 1 - P$$

Q.1 A coin was tossed 400 times and the head turned up 216 times. Test the hypothesis that the coin is unbiased.
 (the prob. of heads is the same as the prob. of tails)

Solⁿ:

probability of getting head = $\frac{1}{2} = 0.5 = P$

H_0 : the coin is unbiased, ($P = \frac{1}{2}$)

vs

H_1 = the coin is not unbiased ($P \neq \frac{1}{2}$)

$n = 400$, X = no of getting head (success)
= 216

$$p = \text{proportion of success} = \frac{X}{n} = \frac{216}{400} = 0.54$$

If prob is not $\frac{1}{2}$ then it is biased or unfair coin.

$$Q = 1 - P = 1 - 0.5 = 0.5$$

Test statistic

$$Z = \frac{p - P}{\sqrt{\frac{PQ}{n}}} = \frac{0.54 - 0.5}{\sqrt{\frac{0.5 \times 0.5}{400}}} = 1.6$$

$$|Z| = 1.6 < Z_{0.05} = 1.96$$

H_0 is accepted

the coin is unbiased.

Q.2 An certain cubical die was thrown 900 times and 5 or 6 was obtained 3240 times. on the assumption of certain throwing, do the data indicate an unbiased die?

Solⁿ

$$n = 9000, X = 3240$$

P = probability of success (i.e. getting 5 or 6)

$$= \frac{2}{6} = \frac{1}{3}$$

$$Q = 1 - P = 1 - \frac{1}{3} = \frac{2}{3}$$

$$p = \frac{X}{n} = \frac{3240}{9000} = 0.36$$

H_0 : Die is unbiased i.e. $P = \frac{1}{3}$

H_1 : Die is biased i.e. $P \neq \frac{1}{3}$

$$Z = \frac{p - P}{\sqrt{\frac{PQ}{n}}} = \frac{0.36 - 0.33}{\sqrt{\frac{\frac{1}{3} \times \frac{2}{3}}{9000}}} = 0.03496$$

$$|Z| = 0.03496 < Z_{0.05} = 1.96$$

$\therefore H_0$ is accepted.

Testing of significance for difference of proportion (large samples)

Let X_1 and X_2 be two samples taken from two different populations. n_1 and n_2 be the size of the samples respectively. If p_1 and p_2 be proportion of success in both samples respectively. Then to test the significant difference between p_1 and p_2 , test statistic is given by

$$Z = \frac{p_1 - p_2}{\sqrt{PQ \left(\frac{1}{n_1} + \frac{1}{n_2} \right)}}$$

$$\text{where } P = \frac{n_1 p_1 + n_2 p_2}{n_1 + n_2} \quad Q = 1 - P$$

Q. In a batch of 500 articles, produced by a machine, 16 articles are found defective. After overhauling the machine, it is found that 3 articles are defective in a batch of 100. Has the machine improved?

Soln.

$$n_1 = 500, \quad n_2 = 100$$

$$p_1 = \frac{16}{500} = 0.032$$

$$p_2 = \frac{3}{100} = 0.03$$

$$P = \frac{p_1 n_1 + p_2 n_2}{n_1 + n_2} = \frac{(0.032)(500) + (0.03 \times 100)}{500 + 100}$$

$$= \frac{19}{600} = 0.032$$

$$Q = 1 - P = 1 - 0.032 = 0.968$$

since we have to test whether machine is proved or not

H_0 : Machine has not improved after overhauling i.e. $P_1 = P_2$

H_1 : $P_1 \neq P_2$

$$\begin{aligned}\text{Under } H_0 \quad Z &= \frac{P_1 - P_2}{\sqrt{PQ\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}} \\ &= \frac{0.032 - 0.03}{\sqrt{0.032 \times 0.968 \times \left(\frac{1}{500} + \frac{1}{100}\right)}} \\ &= 0.104 < 1.645 = Z_{\alpha}\end{aligned}$$

as $|Z| < Z_{\alpha}$.

H_0 is accepted
i.e. Machine is not improved.

Q.2 Random sample of 400 men and 600 women were asked whether they would have a school near their residence. 200 men and 325 women were in favour of proposal. Test the hypothesis that the proportion of men and women in favour of proposal is same at 5% i.e. 0.05.

Solⁿ:

$$n_1 = 400, n_2 = 600$$

$$P_1 = \frac{200}{400} = \frac{1}{2}$$

$$P_2 = \frac{325}{600} = \frac{13}{24}$$

$$P = \frac{P_1 n_1 + P_2 n_2}{n_1 + n_2}$$

$$= \frac{200 + 325}{400 + 600}$$

$$= 0.525$$

$$Q = 1 - P = 0.475$$

H_0 : Proportion of men & women in favour of proposal is same i.e. $P_1 = P_2$

$$H_1: P_1 \neq P_2$$

$$Z = \frac{P_1 - P_2}{\sqrt{PQ \left(\frac{1}{n_1} + \frac{1}{n_2} \right)}}$$

$$= \frac{\frac{1}{2} - \frac{13}{24}}{\sqrt{0.525 \times 0.475 \left(\frac{1}{400} + \frac{1}{600} \right)}}$$

$$= 1.29$$

$$= 1.29$$

$$|Z| = 1.29 < Z_{\alpha} = 1.96$$

H_0 is accepted.